

Introduction to the simplex method

We'll start by explaining the "easy case" of the Simplex Method: when you start with a linear program in standard form where all the right-hand sides of the constraints are non-negative.

Roughly speaking, you turn the LP into a dictionary¹, and then repeatedly pivot to get new dictionaries until at some point the numbers in the dictionary indicate you are done. We'll illustrate the procedure with the following example:

$$\text{Maximize } x_1 + 2x_2 - x_3$$

subject to

$$2x_1 + x_2 + x_3 \leq 14$$

$$4x_1 + 2x_2 + 3x_3 \leq 28$$

$$2x_1 + 5x_2 + 5x_3 \leq 30$$

$$x_1, x_2, x_3 \geq 0$$

Slack variables

The first step will be to introduce **slack variables**, one for each of the constraints (except the positivity constraints). These are simply the difference between the right-hand side and the left-hand side of a constraint. For example, for the first constraint we define a slack variable $x_4 = 14 - 2x_1 - x_2 - x_3$. In terms of this new variable, the first constraint is equivalent simply to $x_4 \geq 0$, the positivity constraint for x_4 .

Introducing these slack variables leads to a linear program equivalent to the original one² but for which all constraints are either *equations* or *positivity constraints*:

$$\text{Maximize } x_1 + 2x_2 - x_3$$

subject to

$$2x_1 + x_2 + x_3 + x_4 = 14$$

$$4x_1 + 2x_2 + 3x_3 + x_5 = 28$$

$$2x_1 + 5x_2 + 5x_3 + x_6 = 30$$

$$x_1, x_2, x_3, x_4, x_5, x_6 \geq 0$$

Dictionaries

Solving for the slack variables we get the initial **dictionary** for our problem:

$$x_4 = 14 - 2x_1 - x_2 - x_3$$

$$x_5 = 28 - 4x_1 - 2x_2 - 3x_3$$

$$x_6 = 30 - 2x_1 - 5x_2 - 5x_3$$

$$z = 0 + x_1 + 2x_2 - x_3$$

The variables on the left, x_4, x_5, x_6 , are the **basic variables** for this dictionary; the set of these variables is called the **basis**. (In the initial dictionary the basic variables are the slack variables, that changes after pivoting.) The rest of the variables are called **non-basic**. Notice that at the bottom we've added a variable z for the objective function.

Each dictionary is a system of equations that is equivalent to the equality constraints from the LP obtained by adding slack variables. In terms of dictionaries, the LP we want to solve is equivalent to the LP obtained by maximizing z subject to the equations in the dictionary and to positivity constraints for all x_i .

The solution associated to a dictionary

To any dictionary there is an **associated solution** of the system of equations: just set all the non-basic variables to zero and compute the values of the basic variables from the dictionary. For the above dictionary the associated solution is

$$x_1 = x_2 = x_3 = 0, x_4 = 14, x_5 = 28, x_6 = 30.$$

Feasibility

The solution associated to a dictionary certainly satisfies all of the constraints that are equations, but may fail to satisfy the positivity constraints. When it does also satisfy the positivity constraints it is said to be a **feasible solution** and the dictionary is called **feasible** too.

Here the initial dictionary is feasible because the LP we started with happened to have all the right-hand sides of constraints non-negative. This needn't always happen and later on we will cover what to do when some constraints have negative right-hand sides; but for now, let's focus on the case when the initial dictionary is feasible, like in this example.

Pivoting

The main idea of the Simplex Method is to go from dictionary to dictionary by exchanging a basic variable for a non-basic one, in such a way that:

- The objective function increases at each step³.
- The dictionary is feasible at every step.

Let's explain how to pick the variables you swap.

Picking a non-basic variable to enter the basis

In the dictionary we have, the z -row is $z = 0 + x_1 + 2x_2 - x_3$. This tells us that for the solution associated to the dictionary, the objective function has the value 0; and it also tells us, for each non-basic variable, whether increasing that variable will increase or decrease z . Since x_1 and x_2 have positive coefficients, increasing them would increase z . On the other hand, increasing x_3 will *decrease* z .

So, to increase z we will choose either x_1 or x_2 to enter the basis.

Either one would be a valid choice and make some progress toward solving the LP, but we'll use a definite rule to pick one:

Standard rule⁴ for picking the entering variable: among the non-basic variables that have a *positive* coefficient in the objective function, choose the one with the *largest* coefficient. If more than one variable is tied for largest coefficient, pick the one with smallest index (e.g., if both x_4 and x_2 had a coefficient of 3, and 3 was the largest, you'd pick x_2).

This is not done for mathematical reasons but for logistical reasons! By having a standard rule for the course, I can make sure that when you solve an LP you don't get crazy numbers, and the course grader has an easier time, too.

Picking a basic variable to exit the basis

We've settled on putting x_2 in the basis and now we need to decide which basic variable to swap it with. The solution associated to our current dictionary has $x_2 = 0$. The idea now is to increase the value of x_2 as much as possible while keeping the resulting dictionary feasible. Let's focus on how the values of the basic variables depend on x_2 (keeping $x_1 = x_3 = 0$):

$$\begin{aligned}x_4 &= 14 - x_2 \\x_5 &= 28 - 2x_2 \\x_6 &= 30 - 5x_2 \\z &= 0 + 2x_2\end{aligned}$$

We see that:

- To keep $x_4 \geq 0$, we can only increase x_2 up to 14.
- To keep $x_5 \geq 0$, we can only increase x_2 up to 14.
- To keep $x_6 \geq 0$, we can only increase x_2 up to 6.

So, to keep all variables non-negative, we can only go up to 6. Since x_6 had the strictest requirement on the entering variable x_2 , we pick x_6 as the exiting variable.

In case two variable are tied for strictest bound on the entering variable we need a tie breaking rule:

Standard rule for picking the exiting variable: In case of ties, pick the variable with the smallest index.

Performing the pivot

So we've decided that x_2 will enter the basis and x_6 will exit. We get the next dictionary by:

- Solving for x_2 in the equation for x_6 .
- Plugging that value for x_2 into the equations for the other basic variables and the equation for the objective function.

This gives us:

$$\begin{aligned}x_2 &= 6 - \frac{2}{5}x_1 - x_3 - \frac{1}{5}x_6 \\x_4 &= 8 - \frac{8}{5}x_1 + \frac{1}{5}x_6 \\x_5 &= 16 - \frac{16}{5}x_1 - x_3 + \frac{2}{5}x_6 \\z &= 12 + \frac{1}{5}x_1 - 3x_3 - \frac{2}{5}x_6\end{aligned}$$

This is a new dictionary that still represents the same LP. The associated solution has $x_1 = x_3 = x_6 = 0, x_2 = 6, x_4 = 8, x_5 = 16$ and $z = 12$. This is a feasible solution so we learn that getting $z = 12$ is possible, and thus the maximum z , which is what we seek, is 12 or larger.

One way to tell you made a mistake

The solution turned out feasible again by virtue of the way we picked the exiting variable. If we had, for example, decided to make x_2 enter but x_4 exit we would have gotten the following dictionary:

$$\begin{array}{rclclcl}
 x_2 & = & 14 & -2x_1 & -x_3 & -x_4 \\
 x_5 & = & 0 & & -x_3 & +2x_4 \\
 x_6 & = & -40 & +8x_1 & & +5x_4 \\
 z & = & 28 & -3x_1 & -3x_3 & -2x_4
 \end{array}$$

The associated solution has $x_6 = -40$ violating the positivity constraint. If that ever happens, you'll know you made a mistake somewhere!

When to stop pivoting

At this point you know how to pivot to increase the objective function while keeping the dictionary feasible. For example, in the last feasible dictionary we can pivot by having x_1 enter. Either x_4 or x_5 would have to exit and our tie-breaking rule has us choose x_4 . After pivoting we get:

$$\begin{array}{rclclcl}
 x_1 & = & 5 & & -\frac{5}{8}x_4 & +\frac{1}{8}x_6 \\
 x_2 & = & 4 & -x_3 & +\frac{1}{4}x_4 & -\frac{1}{4}x_6 \\
 x_5 & = & 0 & -x_3 & +2x_4 & \\
 z & = & 13 & -3x_3 & -\frac{1}{8}x_4 & -\frac{3}{8}x_6
 \end{array}$$

This new dictionary has all of the coefficients in the z -row *negative*, so we can't pick a new entering variable. This means we are done with this problem and $z = 13$ is the maximum value of the objective function. The solution associated to this dictionary, $x_3 = x_4 = x_6 = 0$, $x_1 = 5$, $x_2 = 4$, $x_5 = 0$ is an **optimal solution**.

Why exactly is $z = 13$ the maximum? Clearly the Simplex Method stops here, since there is no way to pick variables for the next pivot according to the rules, but can't there be a sneaky way to increase z , say, by doing a pivot that's against the rules because it *decreases* z , followed by a pivot that makes z even bigger than 13?

There's no need to worry, that can't happen and the dictionary tells us why: we have $z = 13 - 3x_3 - \frac{1}{8}x_4 - \frac{3}{8}x_6$ and each variable has a positivity constraint, so that equation really tells us that $z = 13 - (\text{something non-negative}) \leq 13$.

Additionally we learn that $z = 13$ *only* when $x_3 = x_4 = x_6 = 0$. And having those variable be zero tells us that $x_1 = 5$, $x_2 = 4$, $x_5 = 0$, so that the optimal solution we found is the *only* optimal solution.

¹ There is an alternative way of presenting the Simplex Method using *tableaux* instead of dictionaries. This really just a different way of writing the same calculation. If you've studied linear programming before you are likely to have seen tableaux. It's a good exercise to solve the same LP both ways, so you can see how this dictionary method corresponds to the tableau. If you don't know about tableaux, don't worry about them.

² Notice that this equivalent LP is *not* in standard form.

³ Or, at least, doesn't decrease. Sometimes you are forced to make **degenerate pivots** which don't change the associated solution or the value of the objective function.

⁴ In some course materials you might see this standard rule called "Anstee's rule" instead, named tongue in cheek after Richard Anstee who has taught this course many times. Don't expect non-UBC people to call it Anstee's rule.

— [Omar Antolín Camarena](#)